

NAME

AtomTypesFingerprints.pl - Generate atom types fingerprints for SD files

SYNOPSIS

AtomTypesFingerprints.pl SDFFile(s)...

```
AtomTypesFingerprints.pl [--AromaticityModel AromaticityModelType] [-a, --AtomIdentifierType
AtomicInvariantsAtomTypes | DREIDINGAtomTypes | EStateAtomTypes | MMFF94AtomTypes | SLogPAtomTypes |
SYBYLAtomTypes | TPSAAtomTypes | UFFAtomTypes] [--AtomicInvariantsToUse "AtomicInvariant, AtomicInvariant..." ] [
--FunctionalClassesToUse "FunctionalClass1,FunctionalClass2..." ] [--AtomTypesSetToUse ArbitrarySize | FixedSize] [
--BitsOrder Ascending | Descending] [-b, --BitStringFormat BinaryString | HexadecimalString] [--CompoundID
DataFieldName or LabelPrefixString] [--CompoundIDLabel text] [--CompoundIDMode DataField | MolName |
LabelPrefix | MolNameOrLabelPrefix] [--DataFields "FieldLabel1,FieldLabel2,..." ] [-d, --DataFieldsMode All | Common |
Specify | CompoundID] [-f, --Filter Yes | No] [--FingerprintsLabelMode FingerprintsLabelOnly | FingerprintsLabelWithIDs
] [--FingerprintsLabel text] [-h, --help] [-k, --KeepLargestComponent Yes | No] [-m, --mode AtomTypesCount |
AtomTypesBits] [-i, --IgnoreHydrogens Yes | No] [--OutDelim comma | tab | semicolon] [--output SD |FP |text |all] [
-o, --overwrite] [-q, --quote Yes | No] [-r, --root RootName] [-s, --size number] [--ValuesPrecision number] [
-v, --VectorStringFormat IDsAndValuesString | IDsAndValuesPairsString | ValuesAndIDsString | ValuesAndIDsPairsString] [
-w, --WorkingDir DirName]
```

DESCRIPTION

Generate atom types fingerprints for *SDFFile(s)* and create appropriate SD, FP or CSV/TSV text file(s) containing fingerprints bit-vector or vector strings corresponding to molecular fingerprints.

Multiple SDFFile names are separated by spaces. The valid file extensions are *.sdf* and *.sd*. All other file names are ignored. All the SD files in a current directory can be specified either by **.sdf* or the current directory name.

The current release of MayaChemTools supports generation of atom types fingerprints corresponding to following -a, --AtomIdentifierTypes:

AtomicInvariantsAtomTypes, DREIDINGAtomTypes, EStateAtomTypes,
FunctionalClassAtomTypes, MMFF94AtomTypes, SLogPAtomTypes,
SYBYLAtomTypes, TPSAAtomTypes, UFFAtomTypes

Based on the values specified for -a, --AtomIdentifierType along with other specified parameters such as --AtomicInvariantsToUse and --FunctionalClassesToUse, initial atom types are assigned to all non-hydrogen atoms or all atoms in a molecule

Using the assigned atom types and specified -m, --Mode, one of the following types of fingerprints are generated:

AtomTypesCount - A vector containing count of atom types
AtomTypesBits - A bit vector indicating presence/absence of atom types

For *AtomTypesCount* fingerprints, two types of atom types set size are allowed as value of --AtomTypesSetToUse option:

ArbitrarySize - Corresponds to only atom types detected in molecule
FixedSize - Corresponds to fixed number of atom types previously defined

For *AtomTypesBits* fingerprints, only *FixedSize* atom type set is allowed.

ArbitrarySize corresponds to atom types detected in a molecule where as *FixedSize* implies a fix number of all possible atom types previously defined for a specific -a, --AtomIdentifierType.

Fix number of all possible atom types for supported *AtomIdentifierTypes* in current release of MayaChemTools are:

AtomIdentifier	Total	TotalWithoutHydrogens
DREIDINGAtomTypes	37	34
EStateAtomTypes	109	87
MMFF94AtomTypes	212	171
SLogPAtomTypes	72	67
SYBYLAtomTypes	45	44

```
FingerprintsVector;AtomTypesCount:AtomicInvariantsAtomTypes:ArbitraryS
size;10;NumericalValues;IDsAndValuesString;C.X1.B01.H3 C.X2.B02.H2 C.X2
.B03.H1 C.X3.B03.H1 C.X3.B04 F.X1.B01 N.X2.B02.H1 N.X3.B03 O.X1.B01.H1
O.X1.B02;2 4 14 3 10 1 1 1 3 2
```

```
FingerprintsVector;AtomTypesCount:DREIDINGAtomTypes:ArbitrarySize;8;Nu
mericalValues;IDsAndValuesString;C_2 C_3 C_R F_ N_3 N_R O_2 O_3;2 9 22
1 1 1 2 3
```

```
FingerprintsVector;AtomTypesCount:DREIDINGAtomTypes:FixedSize;34;OrderedNumericalValues;IDsAndValuesString;B_3 B_2 C_3 C_R C_2 C_1 N_3 N_R N_2 N_1 O_3 O_R O_2 O_1 F_ Al3 Si3 P_3 S_3 Cl Ga3 Ge3 As3 Se3 Br In3 Sn 3 Sb3 Te3 I_ Na Ca Fe Zn;0 0 9 22 2 0 1 1 0 0 3 0 2 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

```
FingerprintsBitVector;AtomTypesBits:DREIDINGAtomTypes:FixedSize;34;BinaryString;Ascending;0011101100101010000000000000000000000000
```

```
FingerprintsVector;AtomTypesCount:EStateAtomTypes:ArbitrarySize;11;Num
ericalValues;IDsAndValuesString;aaCH aasC aasN dO dssC sCH3 sF sOH ssC
H2 ssNH sssCH:14 8 1 2 2 2 1 3 4 1 3
```

[illegible][illegible]

```
FingerprintsVector;AtomTypesCount:FunctionalClassAtomTypes:ArbitrarySize;8;NumericalValues;IDsAndValuesString;Ar Ar.HBA HBA HBA.HBD HBD Hal
NI None;22 1 2 3 1 1 1 10
```

```
FingerprintsVector;AtomTypesCount:MMFF94AtomTypes:ArbitrarySize;13;Num
ericalValues;IDsAndValuesString;C5A C5B C=ON CB C00 CR F N5 NC=O O=CN
O=CQ QC=O QR;2 2 1 18 1 9 1 1 1 1 1 1 2
```

[illegible][illegible]

```
FingerprintsBitVector;AtomTypesBits:SLogPAtomTypes:FixedSize;67;Binary
String;Ascending;100010000011001000101110000010001000000100000100000011
00010000000000000000
```

```
FingerprintsVector;AtomTypesCount:SYBYLAtomTypes:ArbitrarySize;9;NumericalValues;IDsAndValuesString;C.2 C.3 C.ar F N.am N.ar 0.2 0.3 0.co2;29 22 1 1 1 1 2 2
```

```
FingerprintsBitVector;AtomTypesBits:SYBYLAtomTypes:FixedSize;44;Binary  
String;Ascending;11010000110011100000100000000000000000000000
```

```
FingerprintsVector;AtomTypesCount:TPSAAtomTypes:FixedSize;47;OrderedNu  
mericalValues;TdsAndValuesString;N1 N2 N3 N4 N5 N6 N7 N8 N9 N10 N11 N1  
2 N13 N14 N15 N16 N17 N18 N19 N20 N21 N22 N23 N24 N25 N26 N O1 O2 O3 O  
4 O5 O6 O S1 S2 S3 S4 S5 S6 S7 S P1 P2 P3 P4 P;0 0 0 0 0 0 0 1 0 0 0 0 0  
0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 2 3 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

```
FingerprintsBitVector;AtomTypesBits:TPSAAtomTypes:FixedSize;47;BinaryString;Ascending;0000000100000000000000001000000000110000000000000000000
```

```
FingerprintsVector;AtomTypesCount:UFFAtomTypes:ArbitrarySize;8;Numeric
alValues;IDsAndValuesString;C_2 C_3 C_R F_ N_3 N_R O_2 O_3;2 9 22 1 1
1 2 3
```

```
FingerprintsVector;AtomTypesCount:UFFAtomTypes;124;OrderedNumerical  
Values;IDSAndValuesString;He4+4 Li Be3+2 B_3 B_2 C_3 C_R C_2 C_1 N_3 N_  
R N_2 N_1 O_3 O_3_z O_R O_2 O_1 F_ Ne4+4 Na Mg3+2 Al3 Si3 P_3+3 P_3+5 P_  
_3+q S_3+2 S_3+4 S_3+6 S_R S_2 Cl Ar4+4 K_ Ca6+2 Sc3+3 Ti3+4 Ti6+4 V_3+  
;0 0 0 0 0 12 0 3 0 3 0 1 0 2 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0  
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

```
FingerprintsVector;AtomTypesCount:UFFAtomTypes:FixedSize;124;OrderedNu
mericalValues;IDsAndValuesString;He+4 Li Be+3 B_3 B_2 C_3 C_R C_2 C_
1 N_3 N_R N_2 N_1 O_3 O_3_z O_R O_2 O_1 F_ Ne+4 Na Mg+2 Al3 Si3 P_3+
3 P_3+5 P_3+q S_3+2 S_3+4 S_3+6 S_R S_2 Cl Ar+4 K_ Ca+2 Sc+3 Ti...;
0 0 0 0 0 9 22 2 0 1 1 0 0 3 0 0 2 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

```
FingerprintsBitVector;AtomTypesBits:UFFAtomTypes:FixedSize;124;BinaryString;Ascending;00000111011001001010000000000000000000000000000000000
```


RA : RingAtom
 FC : FormalCharge
 MN : MassNumber
 SM : SpinMultiplicity

AtomTypes::AtomicInvariantsAtomTypes module is used to assign atomic invariant atom types.

--FunctionalClassesToUse "*FunctionalClass1,FunctionalClass2...*"

This value is used during *FunctionalClassAtomTypes* value of a, --AtomIdentifierType option. It's a list of comma separated valid functional classes.

Possible values for atom functional classes are: *Ar, CA, H, HBA, HBD, Hal, NI, PI, RA*. Default value [Ref 24]: *HBD,HBA,PI,NI,Ar,Hal*.

The functional class abbreviations correspond to:

HBD: HydrogenBondDonor
 HBA: HydrogenBondAcceptor
 PI : PositivelyIonizable
 NI : NegativelyIonizable
 Ar : Aromatic
 Hal : Halogen
 H : Hydrophobic
 RA : RingAtom
 CA : ChainAtom

Functional class atom type specification for an atom corresponds to:

Ar.CA.H.HBA.HBD.Hal.NI.PI.RA

AtomTypes::FunctionalClassAtomTypes module is used to assign functional class atom types. It uses following definitions [Ref 60-61, Ref 65-66]:

HydrogenBondDonor: NH, NH2, OH
 HydrogenBondAcceptor: N[!H], O
 PositivelyIonizable: +, NH2
 NegativelyIonizable: -, C(=O)OH, S(=O)OH, P(=O)OH

--AtomTypesSetToUse *ArbitrarySize* | *FixedSize*

Atom types set size to use during generation of atom types fingerprints.

Possible values for *AtomTypesCount* values of -m, --mode option: *ArbitrarySize* | *FixedSize*; Default value: *ArbitrarySize*.

Possible values for *AtomTypesBits* value of -m, --mode option: *FixedSize*; Default value: *FixedSize*.

FixedSize value is not supported for *AtomicInvariantsAtomTypes* value of -a, --AtomIdentifierType option.

ArbitrarySize corresponds to only atom types detected in molecule; *FixedSize* corresponds to fixed number of previously defined atom types for specified -a, --AtomIdentifierType.

--BitsOrder *Ascending* | *Descending*

Bits order to use during generation of fingerprints bit-vector string for *AtomTypesBits* value of =item

--BitsOrder *Ascending* | *Descending*

Bits order to use during generation of fingerprints bit-vector string for *AtomTypesBits* value of -m, --mode option. Possible values: *Ascending*, *Descending*. Default: *Ascending*.

Ascending bit order which corresponds to first bit in each byte as the lowest bit as opposed to the highest bit.

Internally, bits are stored in *Ascending* order using Perl vec function. Regardless of machine order, big-endian or little-endian, vec function always considers first string byte as the lowest byte and first bit within each byte as the lowest bit.

-b, --BitStringFormat *BinaryString* | *HexadecimalString*

Format of fingerprints bit-vector string data in output SD, FP or CSV/TSV text file(s) specified by --output used during *AtomTypesBits* value of -m, --mode option. Possible values: *BinaryString*, *HexadecimalString*.

Default value: *BinaryString*.

-f, --Filter Yes | No

Specify whether to check and filter compound data in SDFFile(s). Possible values: Yes or No. Default value: Yes.

By default, compound data is checked before calculating fingerprints and compounds containing atom data corresponding to non-element symbols or no atom data are ignored.

--FingerprintsLabelMode *FingerprintsLabelOnly* | *FingerprintsLabelWithIDs*

Specify how fingerprints label is generated in conjunction with --FingerprintsLabel option value: use fingerprints label generated only by --FingerprintsLabel option value or append atom type value IDs to --FingerprintsLabel option value.

Possible values: *FingerprintsLabelOnly* | *FingerprintsLabelWithIDs*. Default value: *FingerprintsLabelOnly*.

This option is only used for *FixedSize* value of -e, --AtomTypesSetToUse option during generation of *AtomTypesCount* fingerprints and ignored for *AtomTypesBits*.

Atom type IDs appended to --FingerprintsLabel value during *FingerprintsLabelWithIDs* values of --FingerprintsLabelMode correspond to fixed number of previously defined atom types.

--FingerprintsLabel text

SD data label or text file column label to use for fingerprints string in output SD or CSV/TSV text file(s) specified by --output. Default value: *AtomTypesFingerprints*.

-h, --help

Print this help message.

-i, --IgnoreHydrogens Yes | No

Ignore hydrogens during fingerprints generation. Possible values: Yes or No. Default value: Yes.

For yes value of -i, --IgnoreHydrogens, any explicit hydrogens are also used for generation of atom type fingerprints; implicit hydrogens are still ignored.

-k, --KeepLargestComponent Yes | No

Generate fingerprints for only the largest component in molecule. Possible values: Yes or No. Default value: Yes.

For molecules containing multiple connected components, fingerprints can be generated in two different ways: use all connected components or just the largest connected component. By default, all atoms except for the largest connected component are deleted before generation of fingerprints.

-m, --mode *AtomTypesCount* | *AtomTypesBits*

Specify type of atom types fingerprints to generate for molecules in SDFFile(s). Possible values: *AtomTypesCount* or *AtomTypesBits*. Default value: *AtomTypesCount*.

For *AtomTypesCount* values of -m, --mode option, a fingerprint vector string is generated. The vector string corresponding to *AtomTypesCount* contains count of atom types.

For *AtomTypesBits* value of -m, --mode option, a fingerprint bit-vector string containing zeros and ones indicating presence or absence of atom types is generated.

For *AtomTypesCount* atom types fingerprints, two types of atom types set size can be specified using -a, --AtomTypesSetToUse option: *ArbitrarySize* or *FixedSize*. *ArbitrarySize* corresponds to only atom types detected in molecule; *FixedSize* corresponds to fixed number of atom types previously defined.

For *AtomTypesBits* atom types fingerprints, only *FixedSize* is allowed.

Combination of -m, --Mode and --AtomTypesSetToUse along with -a, --AtomIdentifierType allows generation of following different atom types fingerprints:

Mode	AtomIdentifierType	AtomTypesSetToUse
AtomTypesCount	AtomicInvariantsAtomTypes	ArbitrarySize [Default]
AtomTypesCount	DREIDINGAtomTypes	ArbitrarySize
AtomTypesCount	DREIDINGAtomTypes	FixedSize
AtomTypesBits	DREIDINGAtomTypes	FixedSize
AtomTypesCount	EStateAtomTypes	ArbitrarySize

AtomTypesCount	EStateAtomTypes	FixedSize
AtomTypesBits	EStateAtomTypes	FixedSize
AtomTypesCount	FunctionalClassAtomTypes	ArbitrarySize
AtomTypesCount	MMFF94AtomTypes	ArbitrarySize
AtomTypesCount	MMFF94AtomTypes	FixedSize
AtomTypesBits	MMFF94AtomTypes	FixedSize
AtomTypesCount	SLogPAtomTypes	ArbitrarySize
AtomTypesCount	SLogPAtomTypes	FixedSize
AtomTypesBits	SLogPAtomTypes	FixedSize
AtomTypesCount	SYBYLAtomTypes	ArbitrarySize
AtomTypesCount	SYBYLAtomTypes	FixedSize
AtomTypesBits	SYBYLAtomTypes	FixedSize
AtomTypesCount	TPSAAAtomTypes	FixedSize
AtomTypesBits	TPSAAAtomTypes	FixedSize
AtomTypesCount	UFFAtomTypes	ArbitrarySize
AtomTypesCount	UFFAtomTypes	FixedSize
AtomTypesBits	UFFAtomTypes	FixedSize

The default is to generate *AtomicInvariantAtomTypes* fingerprints corresponding to *ArbitrarySize* as value of --AtomTypesSetToUse option.

--OutDelim *comma | tab | semicolon*

Delimiter for output CSV/TSV text file(s). Possible values: *comma, tab, or semicolon* Default value: *comma*.

--output *SD | FP | text | all*

Type of output files to generate. Possible values: *SD, FP, text, or all*. Default value: *text*.

-o, --overwrite

Overwrite existing files.

-q, --quote *Yes | No*

Put quote around column values in output CSV/TSV text file(s). Possible values: *Yes or No*. Default value: *Yes*.

-r, --root *RootName*

New file name is generated using the root: <Root>.<Ext>. Default for new file names: <SDFFileName><AtomTypesFP>.<Ext>. The file type determines <Ext> value. The sdf, fpf, csv, and tsv <Ext> values are used for SD, FP, comma/semicolon, and tab delimited text files, respectively. This option is ignored for multiple input files.

-v, --VectorStringFormat *ValuesString | IDsAndValuesString | IDsAndValuesPairsString | ValuesAndIDsString | ValuesAndIDsPairsString*

Format of fingerprints vector string data in output SD, FP or CSV/TSV text file(s) specified by --output used during <AtomTypesCount> value of -m, --mode option. Possible values: *ValuesString, IDsAndValuesString | IDsAndValuesPairsString | ValuesAndIDsString | ValuesAndIDsPairsString*.

Default value during *ArbitrarySize* value of -e, --AtomTypesSetToUse option: *IDsAndValuesString*. Default value during *FixedSize* value of -e, --AtomTypesSetToUse option: *ValuesString*.

Example of *SD* file containing atom types fingerprints string data:

```

... ..
... ..
$$$$
... ..
... ..
... ..
41 44 0 0 0 0 0 0 0 0999 V2000
```



```

-3.3652  1.4499  0.0000 C  0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
... ..
2 3 1 0 0 0 0
... ..
M END
> <CmpdID>
Cmpd1

> <AtomTypesFingerprints>
FingerprintsVector;AtomTypesCount:AtomicInvariantsAtomTypes:ArbitrarySi
ze;10;NumericalValues;IDsAndValuesString;C.X1.B01.H3 C.X2.B02.H2 C.X2.B
03.H1 C.X3.B03.H1 C.X3.B04 F.X1.B01 N.X2.B02.H1 N.X3.B03 O.X1.B01.H1 O.
X1.B02;2 4 14 3 10 1 1 1 3 2

$$$$
... ..
... ..

```

Example of *FP* file containing atom types fingerprints string data:

```

#
# FingerprintsStringType = FingerprintsVector
#
# Description = AtomTypesCount:AtomicInvariantsAtomTypes:ArbitrarySize
# VectorStringFormat = IDsAndValuesString
# VectorValueType = NumericalValues
#
Cmpd1 10;C.X1.B01.H3 C.X2.B02.H2 C.X2.B03.H1 C.X3.B03.H1 C.X3.B04 F...
Cmpd2 9;C.X1.B01.H3 C.X2.B02.H2 C.X3.B03.H1 C.X3.B04 N.X1.B01.H2 N...
... ..
... ..

```

Example of *CSV Text* file atom types containing fingerprints string data:

```

"CompoundID","AtomTypesFingerprints"
"Cmpd1","FingerprintsVector;AtomTypesCount:AtomicInvariantsAtomTypes:Ar
bitrarySize;10;NumericalValues;IDsAndValuesString;C.X1.B01.H3 C.X2.B02.
H2 C.X2.B03.H1 C.X3.B03.H1 C.X3.B04 F.X1.B01 N.X2.B02.H1 N.X3.B03 O.X1.
B01.H1 O.X1.B02;2 4 14 3 10 1 1 1 3 2"
O.X1.B02;3 3 6 3 1 1 2 2 2"
... ..
... ..

```

Examples:

```

FingerprintsVector;AtomTypesCount:ESTateAtomTypes:ArbitrarySize;11;Num
ericalValues;IDsAndValuesString;aaCH aasC aasN d0 dssC sCH3 sF sOH ssC
H2 ssNH sssCH;14 8 1 2 2 2 1 3 4 1 3

```

```

FingerprintsVector;AtomTypesCount:SYBYLAtomTypes:ArbitrarySize;9;Numer
icalValues;IDsAndValuesString;C.2 C.3 C.ar F N.am N.ar O.2 O.3 O.co2;2
9 22 1 1 1 1 2 2

```

```

FingerprintsVector;AtomTypesCount:SYBYLAtomTypes:FixedSize;44;OrderedN
umericalValues;IDsAndValuesString;C.3 C.2 C.1 C.ar C.cat N.3 N.2 N.1 N
.ar N.am N.pl3 N.4 O.3 O.2 O.co2 S.3 S.2 S.o S.o2 P.3 F Cl Br I ANY HA
L HET Li Na Mg Al Si K Ca Cr.th Cr.oh Mn Fe Co.oh Cu Zn Se Mo Sn;9 2 0
22 0 0 0 0 1 1 0 0 2 1 2 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0

```

-w, --WorkingDir *DirName*

Location of working directory. Default: current directory.

EXAMPLES

To generate atomic invariants atom types count fingerprints of arbitrary size in vector string format and create a SampleATFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% AtomTypesFingerprints.pl -r SampleATFP -o Sample.sdf
```

To generate functional class atom types count fingerprints of arbitrary size in vector string format and create a SampleATFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% AtomTypesFingerprints.pl -m AtomTypesCount -a FunctionalClassAtomTypes  
-r SampleATFP -o Sample.sdf
```

To generate E-state atom types count fingerprints of arbitrary size in vector string format and create a SampleATFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% AtomTypesFingerprints.pl -m AtomTypesCount -a EStateAtomTypes  
--AtomTypesSetToUse ArbitrarySize -r SampleATFP -o Sample.sdf
```

To generate E-state atom types count fingerprints of fixed size in vector string with IDsAndValues format and create a SampleATFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% AtomTypesFingerprints.pl -m AtomTypesCount -a EStateAtomTypes  
--AtomTypesSetToUse FixedSize -v IDsAndValuesString  
-r SampleATFP -o Sample.sdf
```

To generate E-state atom types bits fingerprints of fixed size in bit-vector string format and create a SampleATFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% AtomTypesFingerprints.pl -m AtomTypesBits -a EStateAtomTypes  
--AtomTypesSetToUse FixedSize -r SampleATFP -o Sample.sdf
```

To generate MMFF94 atom types count fingerprints of arbitrary size in vector string format and create a SampleATFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% AtomTypesFingerprints.pl -m AtomTypesCount -a MMFF94AtomTypes  
--AtomTypesSetToUse ArbitrarySize -r SampleATFP -o Sample.sdf
```

To generate MMFF94 atom types count fingerprints of fixed size in vector string format and create a SampleATFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% AtomTypesFingerprints.pl -m AtomTypesCount -a MMFF94AtomTypes  
--AtomTypesSetToUse FixedSize -r SampleATFP -o Sample.sdf
```

To generate MMFF94 atom types count fingerprints of fixed size in vector string with IDsAndValues format and create a SampleATFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% AtomTypesFingerprints.pl -m AtomTypesCount -a MMFF94AtomTypes  
--AtomTypesSetToUse FixedSize -v IDsAndValuesString  
-r SampleATFP -o Sample.sdf
```

To generate MMFF94 atom types bits fingerprints of fixed size in bit-vector string format and create a SampleATFP.csv file containing sequential compound IDs along with fingerprints vector strings data, type:

```
% AtomTypesFingerprints.pl -m AtomTypesBits -a MMFF94AtomTypes  
--AtomTypesSetToUse FixedSize -r SampleATFP -o Sample.sdf
```

To generate MMFF94 atom types count fingerprints of arbitrary size in vector string format and create a SampleATFP.csv file containing compound ID from molecule name line along with fingerprints vector strings data, type

```
% AtomTypesFingerprints.pl -m AtomTypesCount -a MMFF94AtomTypes  
--DataFieldsMode CompoundID --CompoundIDMode MolName
```

```
-r SampleATFP -o Sample.sdf
```

To generate MMFF94 atom types count fingerprints of arbitrary size in vector string format and create a SampleATFP.csv file containing compound IDs using specified data field along with fingerprints vector strings data, type:

```
% AtomTypesFingerprints.pl -m AtomTypesCount -a MMFF94AtomTypes
--DataFieldsMode CompoundID --CompoundIDMode DataField --CompoundID
Mol_ID -r SampleATFP -o Sample.sdf
```

To generate MMFF94 atom types count fingerprints of arbitrary size in vector string format and create a SampleATFP.csv file containing compound ID using combination of molecule name line and an explicit compound prefix along with fingerprints vector strings data, type:

```
% AtomTypesFingerprints.pl -m AtomTypesCount -a MMFF94AtomTypes
--DataFieldsMode CompoundID --CompoundIDMode MolnameOrLabelPrefix
--CompoundID Cmpd --CompoundIDLabel MolID -r SampleATFP -o Sample.sdf
```

To generate MMFF94 atom types count fingerprints of arbitrary size in vector string format and create a SampleATFP.csv file containing specific data fields columns along with fingerprints vector strings data, type:

```
% AtomTypesFingerprints.pl -m AtomTypesCount -a MMFF94AtomTypes
--DataFieldsMode Specify --DataFields Mol_ID -r SampleATFP
-o Sample.sdf
```

To generate MMFF94 atom types count fingerprints of arbitrary size in vector string format and create a SampleATFP.csv file containing common data fields columns along with fingerprints vector strings data, type:

```
% AtomTypesFingerprints.pl -m AtomTypesCount -a MMFF94AtomTypes
--DataFieldsMode Common -r SampleATFP -o Sample.sdf
```

To generate MMFF94 atom types count fingerprints of arbitrary size in vector string format and create SampleATFP.sdf, SampleATFP.fpf and SampleATFP.csv files containing all data fields columns in CSV file along with fingerprints vector strings data, type:

```
% AtomTypesFingerprints.pl -m AtomTypesCount -a MMFF94AtomTypes
--DataFieldsMode All --output all -r SampleATFP -o Sample.sdf
```

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SEE ALSO

InfoFingerprintsFiles.pl, SimilarityMatricesFingerprints.pl, AtomNeighborhoodsFingerprints.pl, ExtendedConnectivityFingerprints.pl, MACCSKeysFingerprints.pl, PathLengthFingerprints.pl, TopologicalAtomPairsFingerprints.pl, TopologicalAtomTorsionsFingerprints.pl, TopologicalPharmacophoreAtomPairsFingerprints.pl, TopologicalPharmacophoreAtomTripletsFingerprints.pl

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